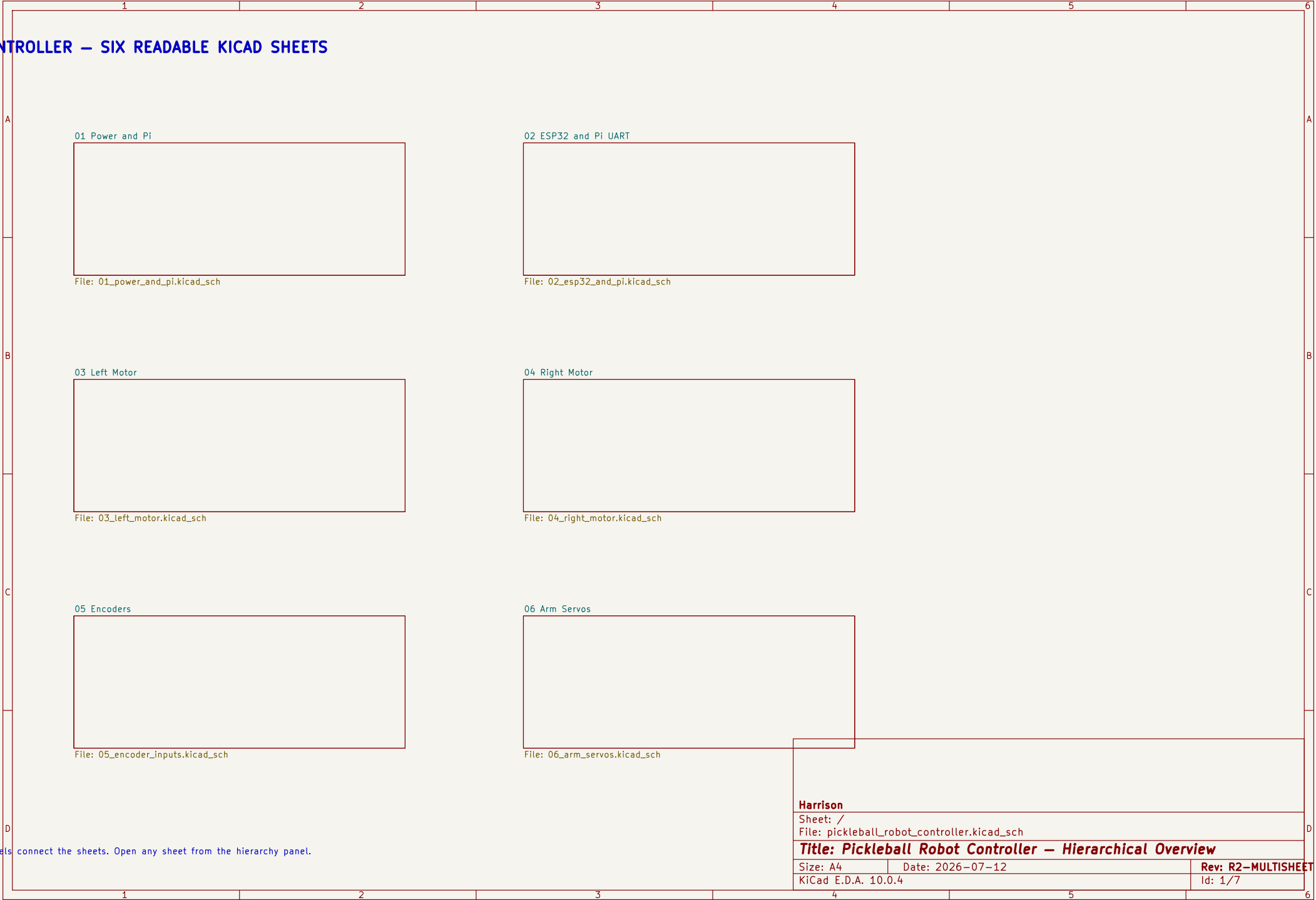


Robot Controller – Six Readable KiCad Sheets



net labels connect the sheets. Open any sheet from the hierarchy panel.

ERY INPUT AND PROTECTION

RASPBERRY PI 5 V / 5 A BUCK

external latching E-stop between F1 and the VBAT_SW distribution point.

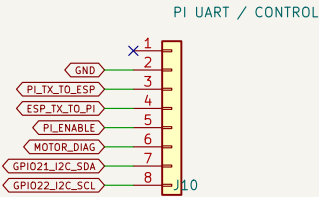
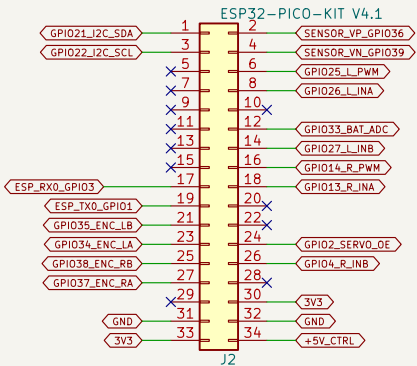
BATTERY VOLTAGE MONITOR

TROL BUCK – ESP32 AND LOGIC ONLY

PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG

CARRIER – CONFIRMED 17.78 mm ROW SPACING

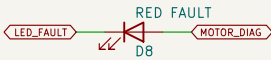
RASPBERRY PI UART / CONTROL – NORMAL 2.54 mm HEADER



J10 pin 1 UNUSED; pin 2 GND; pin 3 Pi TX; pin 4 Pi RX; pins 5-8 optional control/I2C.

Pi GPIO14 physical pin 8 → J10.3; Pi GPIO15 physical pin 10 ← J10.4; Pi GND → J10.2.

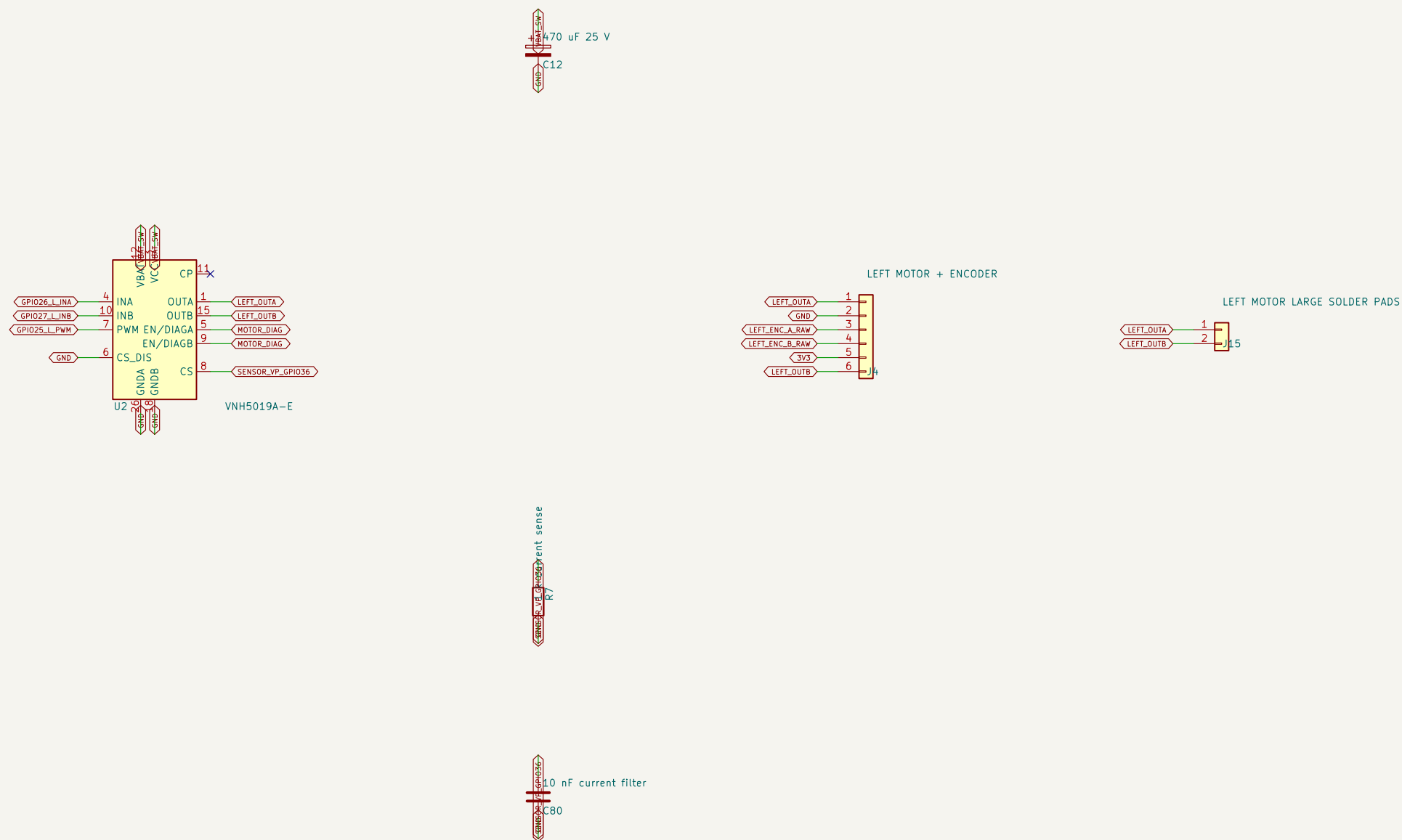
STATUS LEDS



PWR_FLAG



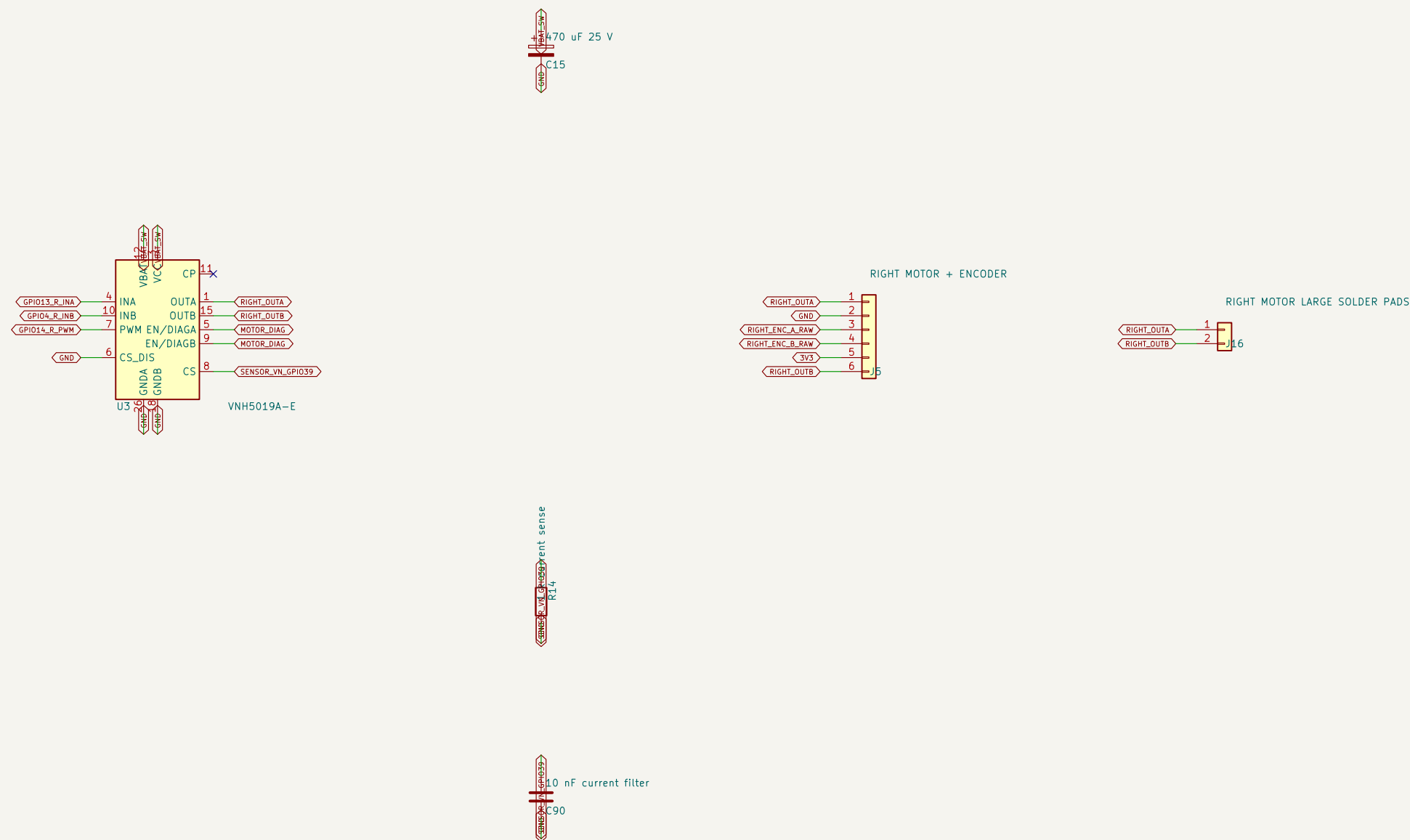
LEFT WHEEL MOTOR DRIVER



< encoder GND; 3 YELLOW encoder A; 4 GREEN encoder B; 5 BLUE encoder +3V3; 6 WHITE motor-.

holes in parallel with pins 1/6 for separately soldered red and white motor-power wires.

RIGHT WHEEL MOTOR DRIVER



< encoder GND; 3 YELLOW encoder A; 4 GREEN encoder B; 5 BLUE encoder +3V3; 6 WHITE motor-.

holes in parallel with pins 1/6 for separately soldered red and white motor-power wires.

SCHMITT-BUFFERED ENCODER INPUTS

A

B

C

D

tors are on the motor-driver sheets; their encoder A/B pins enter here.

F

A

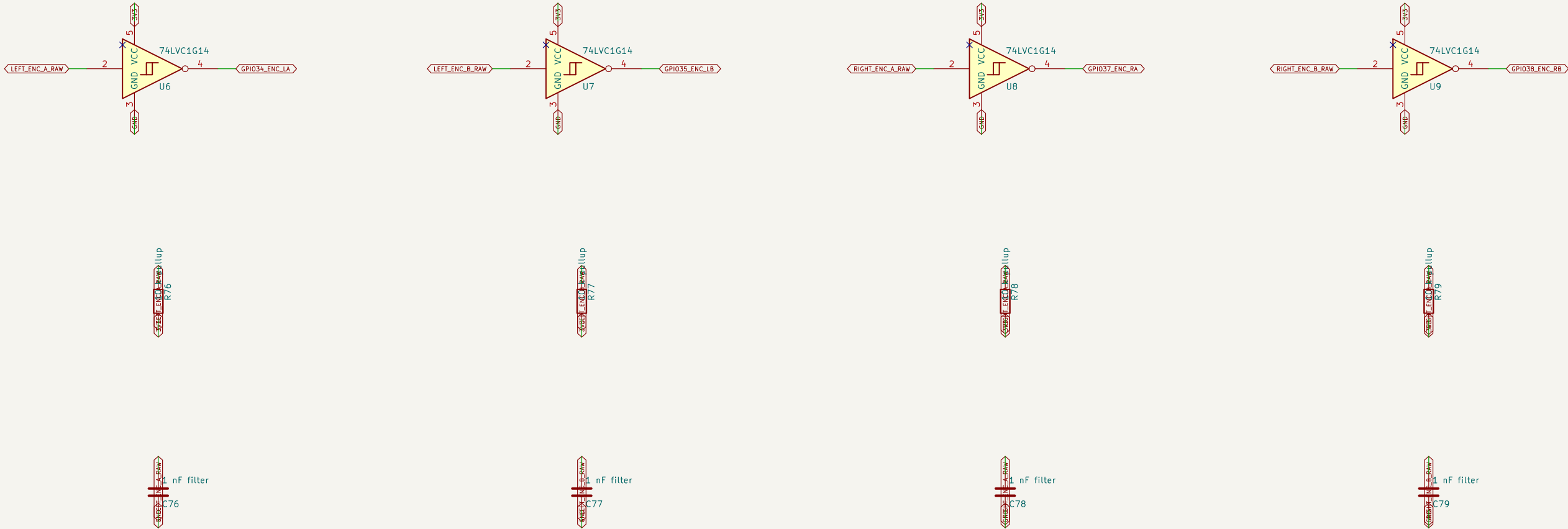
B

C

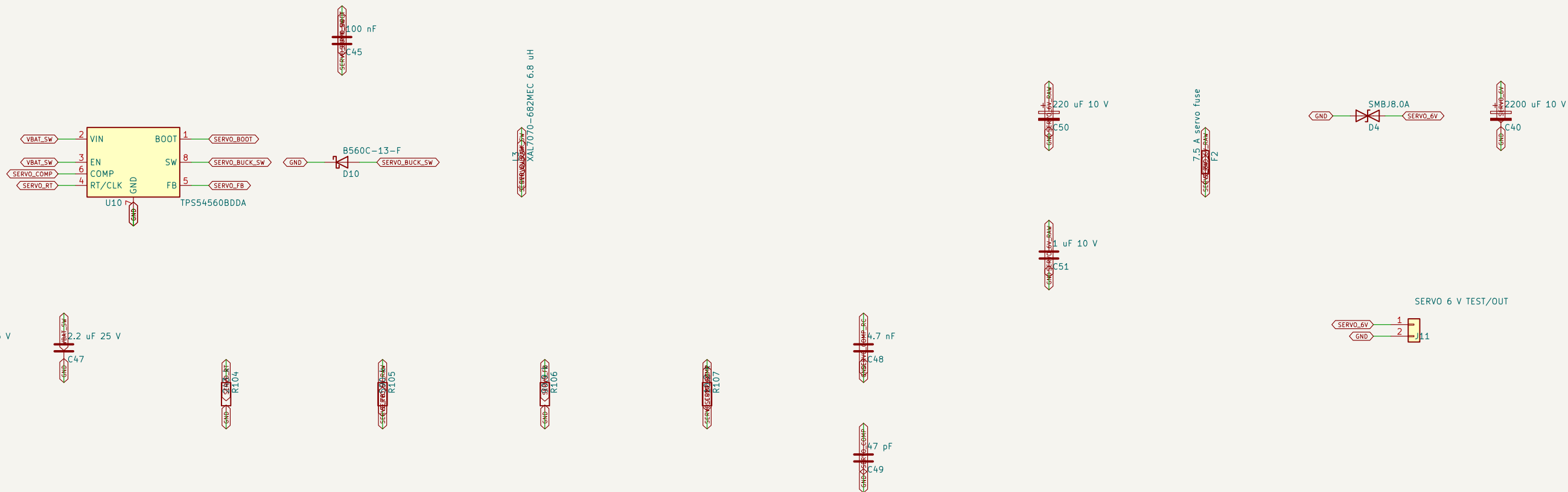
D

E

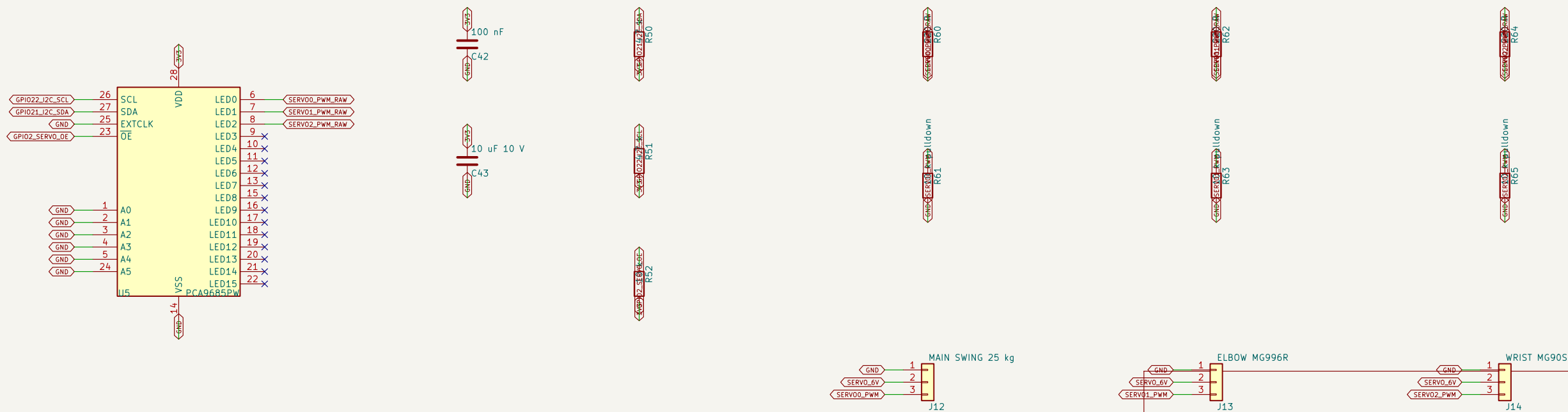
F



6 V / 5 A SERVO BUCK



A9685 SERVO PWM CONTROL



PWR_FLAG

Servo header order: pin 1 GND, pin 2 regulated +6 V, pin 3 PWM.